PH.45320 **FALL 2026**

Static Cosmology

Data Exercise 10

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Topic

Equilibrium of Static Populations

Submission

Submit one ZIP archive containing report.pdf and the requested data table. Record the catalog and calibration versions in the report. The maximum file size is 25 MB.

Problems

1. Solve the first displayed relation used in Equilibrium of Static Populations. State all assumptions and conventions.

$$\partial_t f + v \cdot \nabla_x f - \nabla\Phi \cdot \nabla_v f = C[f] + S[f] - D[f] = 0$$

2. Estimate a nontrivial case using the second relation. Carry units and uncertainty where applicable.

$$N(\tau) = \int R(t-\tau) P_{surv}(\tau) dt$$

3. Analyze the third relation in two limiting regimes and explain the physical meaning of the result.

$$\delta\dot{x} = \mathcal{E} e \delta f$$

Show enough intermediate work that another student can reproduce the calculation.